

GEOLOGIC HAZARD

Earthquake Fault Lines

A mapped fault trace is only one part of earthquake risk: strong shaking can damage homes many miles away.



1 Understand the factor

What it is - and why it matters

An earthquake fault is a fracture or zone of fractures where blocks of the Earth's crust have moved. Active-fault maps identify known or suspected traces, but a line on a map is not a precise prediction of where the next earthquake will occur. Surface rupture is concentrated near the fault; damaging shaking, aftershocks, fires, and utility failures can extend across an entire region. A home's construction era, foundation, soil, slope, and retrofit history often matter as much as straight-line distance to the fault.

How risk can reach a household

- Strong shaking can crack foundations, collapse unbraced walls and chimneys, and overturn unsecured contents.
- Surface rupture can permanently offset land, roads, buried utilities, fences, and structures placed across the trace.
- Broken gas, electrical, water, and sewer lines can create fire, contamination, and loss-of-service hazards.
- Aftershocks can further damage weakened buildings and delay inspections, repairs, and safe re-entry.

What people often miss

- Some faults are concealed, offshore, or mapped with uncertainty; absence of a line does not mean absence of shaking.
- Soft-story construction, unreinforced masonry, raised foundations, and hillside homes have different failure modes.
- Deep sedimentary basins and soft soils can amplify or lengthen shaking even away from a fault trace.
- Standard homeowners insurance usually excludes earthquake shaking and earth movement unless separate coverage is purchased.

? Five checks before buying

1

Check USGS and state geological survey maps, then obtain parcel-level interpretation when the site is near a mapped trace.

2

Review permits and invoices for foundation bolting, cripple-wall bracing, soft-story work, chimney repair, and seismic upgrades.

3

Ask a qualified inspector or structural engineer to evaluate the foundation, lateral bracing, hillside conditions, and additions.

4

Request an earthquake-insurance quote before removing contingencies; compare deductibles, contents, and loss-of-use limits.

5

Confirm emergency access, gas shutoff procedures, water storage, and whether utilities cross known rupture zones.

IMPORTANT DISTINCTION Map proximity is a screening signal - not proof of exposure, damage, or insurability. Confirm the exact feature, current condition, pathway, and property-specific controls.

HEALTH, PROPERTY AND COVERAGE

How this factor can affect a household

FACTOR 01

H Health impacts

- Immediate risks include blunt-force injury, crushing, falls, burns, smoke inhalation, and cuts from glass or debris.
- Displacement, repeated aftershocks, financial loss, and uncertainty can contribute to anxiety, insomnia, and post-traumatic stress.
- Utility failures can interrupt medications, refrigeration, medical devices, clean water, sanitation, and emergency response.
- Children, older adults, people with mobility limitations, and people dependent on powered medical equipment need tailored plans.

P Property impacts

- Structural damage may affect foundations, cripple walls, framing connections, chimneys, masonry, retaining walls, and garages.
- Road, bridge, utility, and neighborhood damage can make an intact home temporarily uninhabitable or difficult to sell.
- Nonstructural losses can be large: water heaters, cabinets, electronics, solar systems, sprinklers, and contents can fail or fall.
- Repairs may trigger current-code requirements, engineering costs, permits, and long delays when contractors are scarce.

I Insurance implications

- Most standard homeowners policies exclude earthquake shaking, ground rupture, settlement, and other earth movement.
- Coverage for contents, masonry veneer, landscaping, pools, detached structures, and loss of use may be limited or optional.
- Separate earthquake policies commonly use percentage deductibles, so the owner may absorb a substantial first layer of loss.
- Fire following an earthquake may be treated differently from shaking damage; the policy wording and cause-of-loss sequence matter.

Possible long-term health implications of buying nearby

- Living in a seismically vulnerable home does not create a continuous toxic exposure, but it creates a low-frequency, high-consequence injury risk.
- Post-event water intrusion, dust, smoke, mold, sanitation failures, and delayed medical care can extend health effects beyond the initial shaking.
- Repeated earthquakes and aftershock cycles can produce chronic stress, sleep disruption, financial strain, and prolonged displacement.
- Retrofitting and preparedness reduce risk; distance to a fault alone cannot determine personal or property safety.

Evidence lens

Evidence is strong for the physical consequences of shaking and structural failure. Parcel-level loss depends on earthquake magnitude, distance, soil, building type, maintenance, and retrofit quality.

Plan more carefully for

Children

Older adults

Pregnancy

Heart/lung conditions

Mobility or power needs

Insurance reality: Availability, eligibility, exclusions, deductibles, and limits vary by state, carrier, property, and cause of loss. Get an address-specific written quote before the purchase contingency expires.

PRACTICAL RISK REDUCTION

Precautions and a real-life example

FACTOR 01

V Verify the actual risk

- Use official fault and shaking maps, not only a real-estate risk badge or a single distance number.
- Keep retrofit permits, engineering reports, foundation photographs, and contractor warranties with the property records.

P Protect the property

- Bolt the sill plate, brace cripple walls, secure the water heater, and correct soft-story or masonry weaknesses where applicable.
- Anchor tall furniture, appliances, solar batteries, and heavy wall objects; use flexible gas connections where permitted.

M Monitor and respond

- Practice Drop, Cover, and Hold On; maintain water, medications, lighting, radios, shoes, and fire extinguishers.
- After shaking, check for gas odor, sparking, water leaks, foundation movement, and unsafe chimneys before re-entry.

I Prepare insurance records

- Compare earthquake policies by deductible, dwelling limit, code-upgrade coverage, contents, and temporary housing.
- Maintain a video inventory and realistic reconstruction estimate; do not assume market value equals rebuild cost.

REAL-LIFE EXAMPLE

1994 Northridge Earthquake

January 17, 1994

A magnitude 6.7 earthquake struck the Los Angeles region. The USGS reports widespread structural and infrastructure damage, thousands of injuries, extensive displacement, and roughly \$40 billion in property damage. Freeways, apartments, hospitals, businesses, and utility systems were affected well beyond the immediate fault area.

Why it matters: The incident showed that a damaging earthquake can occur on a previously less-recognized fault and that building design, soil, retrofit condition, and regional infrastructure strongly influence losses.

[Open official incident record - usgs.gov >](#)

Questions to resolve before making a decision

1. Check USGS and state geological survey maps, then obtain parcel-level interpretation when the site is near a mapped trace.
2. Review permits and invoices for foundation bolting, cripple-wall bracing, soft-story work, chimney repair, and seismic upgrades.
3. Ask a qualified inspector or structural engineer to evaluate the foundation, lateral bracing, hillside conditions, and additions.
4. Request an earthquake-insurance quote before removing contingencies; compare deductibles, contents, and loss-of-use limits.

Official sources and mapping tools

[USGS earthquake maps - usgs.gov >](#)[USGS earthquake hazard and risk FAQ - usgs.gov >](#)[California Department of Insurance guide - insurance.ca.gov >](#)